

Pythagoras' Theorem & Trigonometry

Quick reference sheet · GCSE Maths, Geometry & Measures · keep handy for revision

1. Pythagoras' theorem

For any right-angled triangle, with hypotenuse c (the longest side, opposite the right angle) and shorter sides a and b :

$$a^2 + b^2 = c^2$$

Finding the **hypotenuse**: $c = \sqrt{a^2 + b^2}$ — add the squares, then square root. Example: $a = 6$, $b = 8$
 $\rightarrow c = \sqrt{36+64} = \sqrt{100} = 10$.

Finding a **shorter side**: $a = \sqrt{c^2 - b^2}$ — subtract the squares, then square root. Example: $c = 13$, $b = 5$
 $\rightarrow a = \sqrt{169-25} = \sqrt{144} = 12$.

2. Right-angled trigonometry: SOH CAH TOA

Relative to an angle θ in a right-angled triangle: the **hypotenuse** is the longest side (opposite the right angle); the **opposite** side is across from θ ; the **adjacent** side is next to θ (not the hypotenuse).

Rule	Ratio	Use when you know/want...
SOH	$\sin \theta = \text{opp} \div \text{hyp}$	opposite & hypotenuse
CAH	$\cos \theta = \text{adj} \div \text{hyp}$	adjacent & hypotenuse
TOA	$\tan \theta = \text{opp} \div \text{adj}$	opposite & adjacent

3. Finding a missing side

Rearrange the ratio to make the unknown side the subject. If the unknown is on top (opp or adj), **multiply** by the trig ratio. If the unknown is on the bottom (hyp, or adj in TOA), **divide**.

Example: angle 35° , hypotenuse 9 cm, find the opposite x . $\sin 35^\circ = x \div 9 \rightarrow x = 9 \times \sin 35^\circ = 5.2 \text{ cm}$ (1 d.p.).

4. Finding a missing angle

Use the inverse trig function ($\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$) to find θ from a ratio of two known sides.

Known sides	Formula
opposite & hypotenuse	$\theta = \sin^{-1}(\text{opp} \div \text{hyp})$
adjacent & hypotenuse	$\theta = \cos^{-1}(\text{adj} \div \text{hyp})$
opposite & adjacent	$\theta = \tan^{-1}(\text{opp} \div \text{adj})$

Example: opposite 7 cm, hypotenuse 11 cm. $\sin \theta = 7 \div 11 = 0.6364 \rightarrow \theta = \sin^{-1}(0.6364) = 39.5^\circ$ (1 d.p.).

5. Common mistakes

Mixing up which side is the hypotenuse — it is always the longest side, opposite the right angle, never adjacent to it.

Labelling opposite/adjacent relative to the wrong angle — always check which angle is θ first.

Forgetting to use the inverse function ($\sin^{-1}$ etc.) when finding an angle, rather than a side.

Calculator left in radian mode instead of degree mode — always check before using $\sin/\cos/\tan$.

Adding instead of subtracting (or vice versa) when rearranging $a^2 + b^2 = c^2$ to find a shorter side.